Qin Yang

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Education

New York University *2021 Sep - 2023 May*

- Master's Degree in Data Science
- Relevant Courses: **Machine Learning**, Applied ML in Finance, **NLP and Computational Semantics**, **Deep Learning, Time Series Analysis**, **Probability & Statistics, Big Data**

Shandong University *2014 Sep - 2021 May*

- Master's Degree in Economics/Bachelor's Degree in Management Science & Engineering
- Relevant Courses: Advanced Micro & Macro Econ, Econometrics, Finance, **Calculus, Linear Algebra**, **Data Principles and Applications, Python and Data Analysis**, **Database Management**

Experience

Data Scientist, Marketing @Tractor Supply Co. *Brentwood, TN | 2025 Jan - Present*

- Perform customer segmentation and behavioral analysis to identify high-value audiences for targeted campaigns.
- Build and maintain data pipelines to integrate and process large marketing datasets using **Snowflake** and **Azure Data Factory**.
- Design and manage **Power BI** dashboards tracking 20+ KPIs (Conversion Rate, CTR, Sales, Churns, Customer Acquisition Cost, CPM, Gross Margin, Impressions, etc.) to inform strategic decisions.
- Partner with cross-functional teams to translate data insights into marketing optimizations.

Data Scientist, Global Automation & Engineering @Assurant *Nashville, TN | 2023 Feb - Jan 2025*

- **ML & MLOps:** Develop **end-to-end ML** solutions to enhance overall performance in manufacturing workflows. This involves translating complex business requirements into scalable and data-driven ML applications, developing high-performing ML models, building and optimizing **CI/CD** pipelines to streamline the **deployment** and **maintenance** of ML solutions in **production environments** using tools such as **Docker**, **Azure ML**, **Mlflow**, **scikit-learn**, and **Git**. At Assurant, I successfully implemented an algorithm that integrates **NLP** and **CV** to automate USB debug mode on Android phones, resulting in a $300k annual reduction in labor costs and generating additional revenue through external sales.
- **Data Analysis & Reporting**: Conduct data mining and data analysis to support strategic decision-making processes for **VP-level** leadership. This involves utilizing tools such as **Python**, **Pandas/Numpy**, **SQL**, **PySpark** to perform in-depth data analysis on large-scale datasets, developing **real-time data visualization apps** that not only streamline repetitive data analysis and cleaning tasks but also effectively translate complex insights into actionable recommendations through dynamic and intuitive graphical representations. At Assurant, I led the development of real-time data apps that integrate, analyze, and visualize warehouse data from various sources (e.g., **databases**, **APIs**, and **IoT** devices) using tools like **Streamlit, Django, JS, CSS,** and **HTML**. These apps now empower users, including warehouse operators and executives, to access critical insights quickly, facilitating informed decision-making across the organization.
- **Collaboration & Communication**: Lead cross-functional brainstorming sessions to address challenges using data-driven solutions, fostering a culture of teamwork by encouraging open dialogue and sharing of ideas across departments. I also utilize project management tools (e.g., **JIRA**) to track progress, assign tasks, and maintain clear communication across teams, ensuring timely delivery of data initiatives that align with organizational goals.

Research Assistant (Automatic Speech Recognition, ASR), Center for Data Science@NYU *New York, NY | 2022 Sep - 2023 May*

- **Data Preprocessing**: Employed various data cleaning and augmentation techniques to transform raw audio data (375 hours) into high-quality inputs for speech recognition models for effective model training and evaluation.
- **NLP & ASR**: Fine-tuned an ASR model on both labeled and pseudolabeled Holocaust testimony audio data using PyTorch, followed by the inclusion of a word-based Language Model in final search, drove down the Word Error Rate to 13.5%, a huge gain in performance. Weights & Bias (W&B) and High-Performance Computing (HPC) are utilized to track experiments and accelerate the training process respectively.
- **Academic Writing**: Honed my skills in articulating complex ideas clearly and collaborating on research initiatives, as demonstrated by my publication at the 2023 Interspeech conference.

Data Science Intern, Digital Transformation@Campana & Schott *New York, NY | 2022 May - 2022 Sep*

- **ETL**: Implemented robust ETL (Extract, Transform, Load) processes to streamline data collection and preparation. This included extracting data from various sources (APIs, DBs), transforming it into a usable format by cleaning and aggregating it, and loading it into centralized databases for further analysis or modeling.
- **AI/ML**: Utilized advanced AI/ML techniques to address the complex challenges faced by pharmaceutical clients. This involved extracting meaningful insights from **unstructured** patient comments using **NLP** techniques such as **Named Entity Recognition (NER)** and **sentiment analysis**, employing **Time Series** and **Recurrent Neural Network (RNN)** models, such as **Long Short-Term Memory (LSTM)** networks and **N-HiTS** to capture historical sales fluctuation patterns, facilitating better strategic planning and resource allocation.
- **Story Telling**: Developed comprehensive reporting frameworks to visualize and communicate insights to stakeholders effectively. This involved using tools such as **Power BI** to create interactive dashboards and reports that highlighted KPIs and trends, enabling stakeholders to make informed decisions based on real-time data analysis and facilitating ongoing monitoring of business metrics.

Data Science Intern, R&D@iFLYTEK *China | 2019 May - 2019 Sep*

- **Data Preprocessing**: Executed noise removal, audio file normalization, and data consistency checks to enhance large audio datasets, leading to more effective modeling for **Automatic Speech Recognition (ASR).** This involved streamlining preprocessing tasks by developing automated scripts using **Python** and boosting the model's robustness and generalizability across various audio inputs by applying data augmentation techniques.
- **Cross-Functional Teamwork**: Collaborated closely with researchers and engineers to analyze data patterns, identify challenges, and implement robust preprocessing methodologies, resulting in improved accuracy and performance of speech recognition models.

Skills & Tools

- **Scripting Languages**: Python, SQL
- **Machine Learning**: Regression, Classification, & Clustering, Recommendation Systems (e.g., RNN-based, DNN-based, Popularity-based, Collaborative Filtering, etc.)
- **AI**: NLP (Text Classification, Sentiment Analysis, NER, Chatbot, RAG), ASR
- **Visualization**: Power BI, Plotly, Dash
- **DBs**: PostgreSQL, MongoDB, MySQL, Vector DB, Azure Cosmos DB
- **Other**: Time Series Analysis, Langchain, Linux, Multithreading, Multiprocessing, Office, HTML, CSS, Git, FastAPI, PyTorch, Tensorflow, Mlflow, Flask, Django, Azure ML, Streamlit, Docker, Data Analytics, RESTful API, CI/CD, Version Control, ETL, JIRA, Snowflake
- **Soft skills**: Stakeholder Engagement, Interdisciplinary Communication, Collaborative Problem Solving, Project Management, Feedback & Iteration, Cross-team Collaboration, Advanced Analytics, Supply Chain